FACTORY REALLOCATION & SHIPPING OPTIMIZATION

A Decision-Intelligence Framework for Product-to-Factory Assignment and Distribution Analysis at Nassau Candy Distributor

PROJECT RESEARCH PAPER

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Domain	Data Analytics · Decision Support · Operations Research
Tools	Python, Pandas, NumPy, Plotly, Streamlit, geospatial analysis where validated
Dataset	Nassau Candy Distributor transaction dataset; 10,194 rows and 18 columns as inspected during project work
Status	Methodology and preliminary analysis; optimization outcomes require validation

Important scope note

This paper separates observed dataset findings from proposed methods. It does not claim measured shipping-cost savings or a validated optimal factory assignment. The inspected order and ship dates produce implausibly long lead times and must be confirmed with the data owner before date-based performance claims are made.

Prepared for project review and internship submission

Abstract

Distribution businesses must balance customer service, shipping expense, product availability, and profitability. This project proposes a decision-intelligence workflow for analysing sales and gross profit and exploring whether product-to-factory assignments could be improved for the Nassau Candy Distributor dataset. The inspected data contains 10,194 transaction rows and 18 columns, including order and ship dates, shipping mode, customer geography, product, sales, units, gross profit, and cost. The workflow combines data-quality checks, descriptive analytics, product and regional profitability analysis, candidate factory-location data, geographic distance calculations, and a transparent what-if reallocation simulator. A Streamlit interface makes the analysis accessible to non-technical reviewers. During validation, order dates fell in 2024–2025 while ship dates fell in 2026–2030, producing lead times of approximately 904–1,642 days. Lead-time prediction is therefore not treated as reliable until source dates are verified. Factory assignments and cost improvements are hypotheses to test—not proven savings—because factory capacity, actual routing, shipping tariffs, and assignment constraints are not available in the inspected schema. The contribution is a reproducible analytical framework with validation safeguards and a path toward evidence-based operational recommendations.

Keywords: decision intelligence, supply chain analytics, factory allocation, shipping optimization, profitability analysis, data quality, Streamlit

1. Introduction

Distribution businesses need more than historical reporting: decision-makers need to understand which products and regions contribute to revenue and gross profit, how shipping choices vary, and whether alternative fulfilment locations could improve performance. Descriptive analytics can identify where value is generated; scenario analysis can test possible changes under explicit assumptions.

This project uses the Nassau Candy Distributor transaction dataset as a case study. The intended output is a practical dashboard and documented workflow that moves from data inspection to profitability analysis and, where evidence permits, factory-assignment scenarios. The emphasis is on transparent, reproducible analysis rather than unsupported claims of operational improvement.

1.1 Problem statement

The project needs a structured way to inspect sales and profitability and evaluate candidate factory assignments. However, the available transaction schema does not include an observed factory assignment, shipment distance, freight charge, delivery service level, or factory capacity. Date validation also revealed implausible order-to-ship intervals. These gaps prevent a defensible claim that a particular reallocation is optimal. The project is therefore framed as a decision-support prototype and a specification of the additional evidence needed for operational optimization.

1.2 Objectives

- Profile the dataset and document data-quality risks before modelling.
- Summarize sales, cost, gross profit, units, and gross-profit margin by region, division, and product.
- Inspect shipping-mode patterns without treating shipping mode as proof of cost or delivery performance.
- Prepare candidate factory and product-mapping data as explicit, reviewable inputs.
- Estimate geographic distance only when customer and factory coordinates are valid and the method is disclosed.
- Provide a scenario interface for comparing assignments under stated assumptions.
- Communicate limitations so the dashboard is not mistaken for a production-ready optimizer.

2. Dataset and Data Understanding

The inspected dataset contains 10,194 rows and 18 columns. Fields include transaction identifiers, order and ship dates, shipping mode, customer identifier and location, division, region, product identifier and name, sales, units, gross profit, and cost. The schema supports descriptive commercial analysis, but it does not directly identify the factory that fulfilled each order or the actual freight charge.

Attribute group	Available fields	Analytical use
Transaction	Row ID, Order ID	Record tracking and order grouping

Attribute group	Available fields	Analytical use
Dates and shipping	Order Date, Ship Date, Ship Mode	Date-quality checks; descriptive mode analysis
Customer geography	Country/Region, City, State/Province, Postal Code, Region	Regional summaries; geospatial enrichment if reliable
Product and organization	Division, Product ID, Product Name	Product/division aggregation and candidate mapping
Commercial measures	Sales, Units, Gross Profit, Cost	Profitability and scenario inputs

2.1 Data preparation

- Preserve an untouched copy of the raw CSV.
- Check column names, data types, row count, missing values, duplicates, and numeric conversion failures.
- Parse dates using the source's observed day-month-year representation, then inspect date ranges and order-to-ship differences.
- Calculate gross-profit margin as gross profit divided by sales only when sales are positive.
- Keep proposed factory mappings separate and label them as assumptions until confirmed by an authoritative source.

2.2 Critical date-quality finding

Date validation during project work found Order Date values spanning 2024–2025 and Ship Date values spanning 2026–2030. Calculated order-to-ship intervals were approximately 904–1,642 days, with a reported mean of about 1,320.84 days and median of about 1,274 days. The Order ID year component also did not match the parsed Order Date year for the inspected rows. This may reflect synthetic, transformed, or incorrectly assembled data; the cause cannot be established from the current file alone.

Do not train or report a lead-time prediction model on these dates until the source is clarified. Do not silently shift dates, infer replacement dates from Order IDs, or delete rows just to make values plausible. Obtain a corrected source file or written clarification first.

3. Analytical Methodology

3.1 Descriptive sales and profitability analysis

The first analytical layer aggregates commercial measures by region, division, and product. For a group, total sales and total gross profit are sums across its transactions. Gross-profit margin is total gross profit divided by total sales, multiplied by 100, when sales are positive. Calculating the ratio from group totals avoids treating every transaction as equally important when computing group-level margin.

Group gross-profit margin (%) = 100 × Σ Gross Profit / Σ Sales

Product rankings require care: high total sales do not necessarily mean the highest profitability, and a high margin on low volume may not produce the largest total gross profit. Reporting both contribution and margin supports a balanced interpretation.

3.2 Shipping-mode analysis

Shipping mode can be profiled by order count, sales, units, and gross profit. However, shipping mode alone does not establish actual shipping expense or delivery speed. Conclusions about the cheapest or fastest mode require reliable freight charges, service-level definitions, and valid shipment dates. Outputs should be labelled descriptive unless these measures are available.

3.3 Factory-location and distance analysis

If customer and factory coordinates are verified, great-circle distance can be estimated using the Haversine formula. This is a geographic proxy between two points, not road distance, carrier routing distance, or freight cost. Invalid, missing, or ambiguous geocodes must be flagged rather than silently accepted.

3.4 Scenario-based reallocation

The reallocation component should compare an explicitly defined baseline with candidate alternatives. Each scenario needs a product-to-factory mapping, distance source, estimated freight-cost formula, factory-capacity assumptions, and restrictions on which products each factory can produce. Each alternative should be evaluated against the same orders and cost assumptions as the baseline.

Component	Required definition	Current limitation
Baseline assignment	Observed factory per product/order	Not in transaction schema; mapping unverified
Distance	Verified coordinates or route distance	Straight-line distance is only a proxy
Shipping cost	Freight invoices or disclosed cost assumptions	No observed freight charge in listed columns
Factory constraints	Capacity, product eligibility, inventory limits	Not represented in current transaction data
Objective	Minimize cost/distance while preserving service and feasibility	Needs stakeholder-defined constraints

4. Dashboard and Implementation

A Streamlit dashboard serves as the user-facing layer. The prototype organizes the analysis into overview, product analysis, factory assignments, and data explorer views. Plotly charts support interactive regional and product comparisons; filtered data can be downloaded for inspection.

- Executive overview: sales, gross profit, cost, units, and aggregate margin for selected filters.
- Regional analysis: sales and gross profit by region.
- Product analysis: product-level sales, gross profit, units, and ranking.
- Factory assignments: proposed mapping and candidate factory locations with an explicit assumption warning.
- Data explorer: filtered transaction table and CSV download.

The dashboard is a decision-support prototype, not a production logistics system. A working interface does not validate source data, prove factory feasibility, or demonstrate realized cost savings.

5. Evaluation Plan

The current version prioritizes analytical validity over an unsupported predictive score. Once source dates and operational inputs are corrected, evaluation should be carried out in separate layers:

- Data validation: reconcile date ranges, order identifiers, units, sales, cost, and gross-profit definitions against source documentation.
- Descriptive analysis: independently verify group totals and margins using spreadsheet or SQL checks.
- Distance validation: sample-check geocoded points and compare straight-line distances with road-route distances where relevant.
- Cost model validation: calibrate estimated shipping cost against historical freight invoices or a documented carrier tariff.
- Scenario evaluation: compare baseline and alternatives on the same order set; report estimated cost, distance, service constraints, and profit impact.
- Sensitivity analysis: vary freight rates, distance limits, and capacity assumptions to test recommendation stability.
- User review: have an operations stakeholder validate product-factory eligibility and practicality.

6. Preliminary Findings and Interpretation

The work to date supports three defensible conclusions. First, the dataset contains commercial dimensions for meaningful sales and gross-profit summaries. Second, date fields fail basic plausibility checks for shipment lead-time modelling and remain unresolved. Third, the transaction schema alone is insufficient to prove an optimal factory reallocation because it lacks observed factory assignment, freight cost, and capacity constraints.

Product and regional profitability outputs can still support descriptive decision-making by showing where revenue and gross profit are concentrated. Any specific product-to-factory recommendation must remain a candidate scenario until mapping,

distance, costs, and operational feasibility are validated.

7. Limitations and Risk Controls

- No verified observed factory-assignment column is available in the inspected schema.
- No actual freight charges or delivery-service metrics are present in the listed columns.
- Order and ship dates yield implausibly long intervals; lead-time prediction is suspended pending clarification.
- Proposed product-to-factory mapping may be an assumption and requires authoritative validation.
- Factory coordinates and customer geocodes require quality checks; straight-line distance is not road distance.
- Factory capacity, production capability, inventory, customer commitments, and service constraints are not currently modelled.
- Scenario estimates must not be described as realized savings or production recommendations without validation.

8. Ethical and Reproducibility Considerations

Preserve raw data, document transformations, and make assumptions visible in code and dashboard text. If customer locations or identifiers are sensitive, remove, anonymize, or exclude them before public repository publication. Check dataset licensing and internship rules before sharing source data. Results should be reproducible from documented dependencies and input files; any future model should use a justified train/test strategy and leakage checks.

9. Conclusion and Future Work

This project establishes a practical foundation for a factory-reallocation decision-support workflow using transaction-level sales and profitability data. It combines data-quality checks, descriptive analytics, geospatial distance analysis, a transparent scenario concept, and a Streamlit interface. Its most important methodological decision is to stop unreliable lead-time modelling rather than treat implausible dates as valid observations.

The next phase should obtain a validated dataset and confirm current factory assignments for each product or order. It should then add actual freight costs or a documented cost model, product-factory eligibility, factory capacity, and service constraints. With these inputs, the project can progress from a what-if tool to a defensible optimization model evaluated against a baseline and reviewed by an operations stakeholder.

10. Reproducibility Checklist

- Keep the raw dataset unchanged and record its source/version.
- Run date-quality checks before computing or modelling lead time.
- Document all assumptions in the product-factory mapping file.
- Validate factory coordinates and customer geocodes.
- Separate descriptive facts from scenario estimates in charts and text.
- Include dependency versions in requirements.txt and usage instructions in README.md.
- Do not publish restricted datasets, personal information, or confidential internship materials.

References and Technical Resources

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Appendix A — Suggested repository structure

```
nassau-factory-optimization/
■■■ app.py
■■■ requirements.txt
■■■ README.md
■■■ notebooks/
■ ■■■ 05_Date_Validation.ipynb
■ ■■■ 06_Sales_Profitability_Analysis.ipynb
■ ■■■ 07_Factory_Location_Analysis.ipynb
■ ■■■ 08_Shipping_Distance_Analysis.ipynb
■ ■■■ 09_Factory_Reallocation_Simulation.ipynb
■■■ data/ (share only if permitted)
■■■ reports/research_paper.pdf
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Appendix B — Suggested future optimization formulation

Let x(p,f) equal 1 when product p is assigned to factory f, and 0 otherwise. A future model could minimize estimated total fulfilment cost Σ[p,f] demand(p) × estimated_cost(p,f) × x(p,f), subject to each product being assigned to an eligible factory, factory capacity limits, product-factory eligibility, and service-level constraints. This is a proposed formulation only; demand, eligibility, capacity, and cost inputs must be verified before solving or interpreting results.